Representation joins: paper and implementation boundary

Research implementation report — 21 October 2026

Status. Bounded native reference implementation, with an index-calculus relation adapter. No timing or asymptotic improvement is claimed. The full fast backend and production collector integration remain open.

The paper. Delaplace, Fouque, Kirchner and May, *Solving ECDLP over $\mathbb{F}_{p^\ell}$ with Pre-computation via Representation Technique*, 2020, `https://inria.hal.science/hal-02427655v1`. For $Q = kP$, represent $k = k_{11} + k_{12} + k_{21} + k_{22} \pmod r$ in many ways. Join two pairs of lists, retain intermediate points with $x \in \mathbb{F}_p$, and match full points. The match determines a scalar candidate, checked by replaying $kP = Q$. The paper uses polynomial multipoint evaluation to avoid testing every pair. Its quadratic-field setup is approximately $p^{6/5}$, online work and memory $p^{4/5}$, under its heuristics and suppressed overheads. This does not improve the prior Bernstein–Lange precomputation tradeoff.

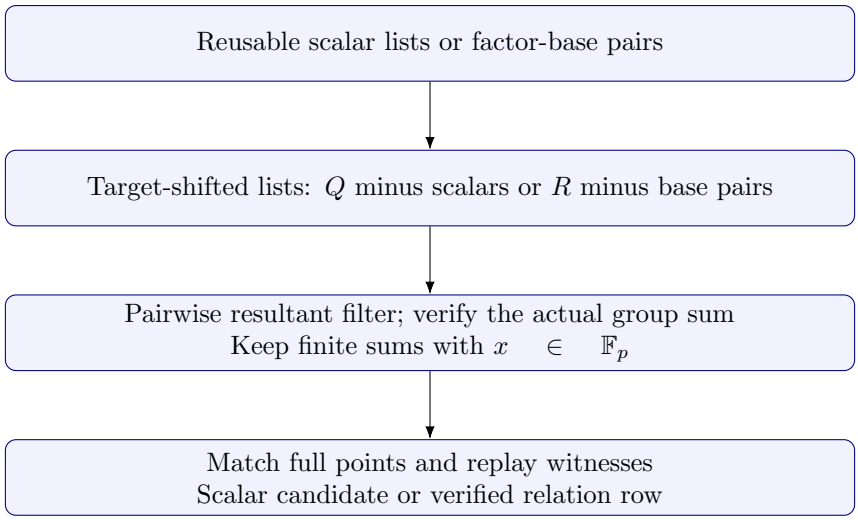


Diagram: implemented reference stages, not measured costs. The SVG companion shows the left and right inputs separately.

Connection to index calculus. The adapter emits witnesses for $aP + bQ = B_i + B_j + B_k + B_l$ with the filtered intermediate. These give rows $a + bk = \sum_j c_j \log_P(B_j) \pmod r$ after the caller validates a common prime-order subgroup. This is a restricted decomposition oracle; its missing unfiltered witnesses must not be counted as complete PDP coverage.

What is implemented. C++17 quadratic-field arithmetic, validated curves, direct and resultant-filter joins, reusable four-list setup, target matching, and coefficient rows retaining witness multiplicities. The resultant is a necessary test, followed by exact addition. Zero determinants can be false positives; infinity and doubling take exact paths. Complexity is still pairwise; bounded membership enumeration costs $O(r)$.

Checks and limitations. Three fixtures over $p = 5, 7, 11$ compare complete join outputs with direct addition; positive subgroups have orders 7 and 137, with all 144 targets checked. The $p = 7$ order-3 subgroup is a zero-yield negative control. Four-index enumeration checks the PDP witness sets; it shares arithmetic with the candidate. See `records/final-check.txt` and `records/sanitizer.txt` for exact results. Earlier fixture failures are retained. These are unit tests, not independent-host replay or performance experiments.

Open work. Symbolic product polynomials, Kedlaya–Umans evaluation, optimized scalar partitions, general extension fields, production collector wiring, rank/LA/descent, and matched end-to-end costs. Prime fields have no proper subfield for this filter. Cargo and Conductor were unavailable locally. No leaderboard, benchmark index, or curve/isogeny graph is promoted.